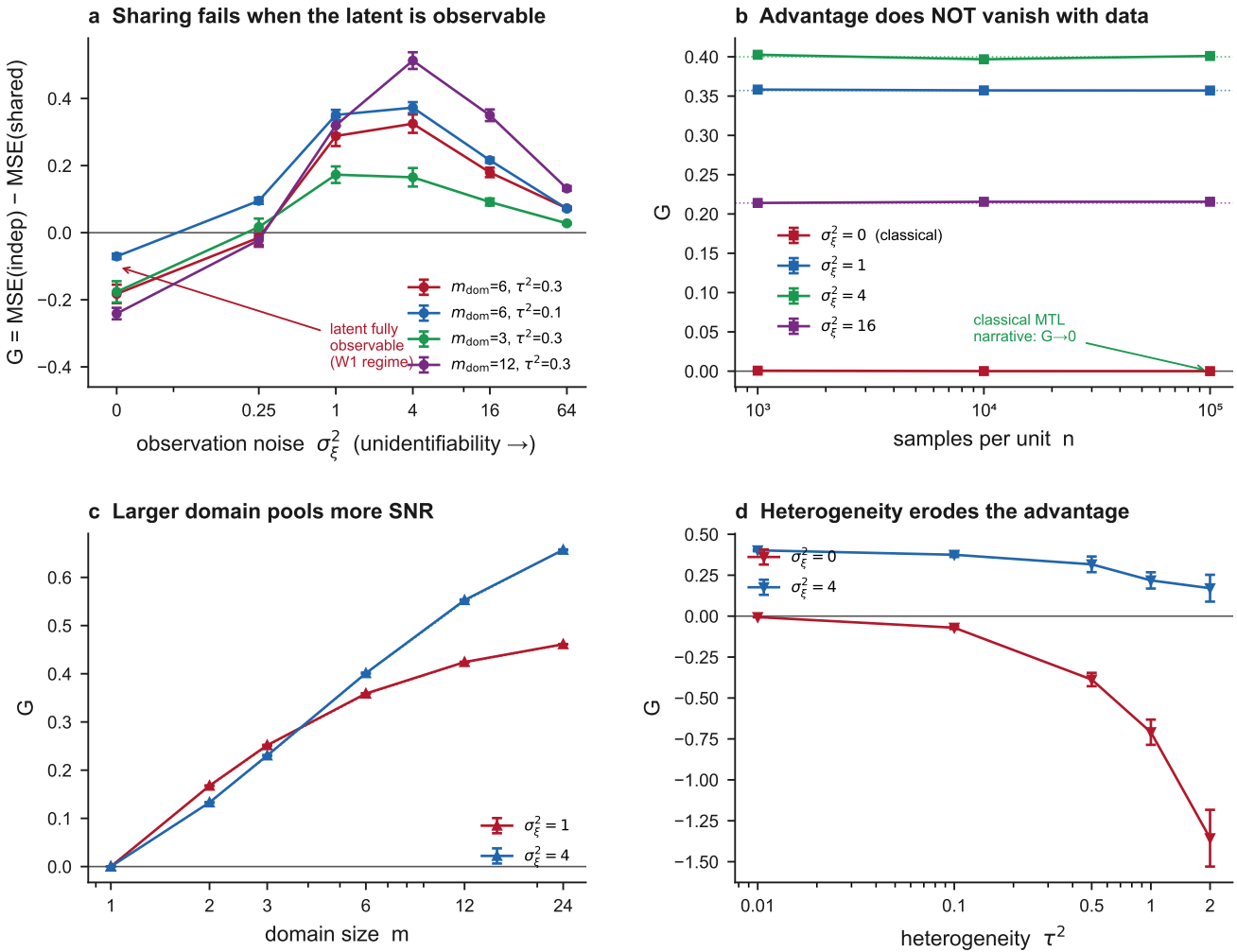
a Sharing fails when the latent is observable
G = MSE(indep) − MSE(shared)
0.4
0.2
0.0
−0.2
−0.4
0
0.25
1
4
16
64
observation noise σ_ξ^2 (unidentifiability →)
latent fully observable (W1 regime)
m_{dom}=6, τ^2=0.3
m_{dom}=6, τ^2=0.1
m_{dom}=3, τ^2=0.3
m_{dom}=12, τ^2=0.3
b Advantage does NOT vanish with data
G
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$\sigma_\xi^2 = 0$ (classical)
$\sigma_\xi^2 = 1$
$\sigma_\xi^2 = 4$
$\sigma_\xi^2 = 16$
classical MTL narrative: G→0
10^3
10^4
10^5
samples per unit n
c Larger domain pools more SNR
G
0.6
0.5
0.4
0.3
0.2
0.1
0.0
$\sigma_\xi^2 = 1$
$\sigma_\xi^2 = 4$
1
2
3
6
12
24
domain size m
d Heterogeneity erodes the advantage
G
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
$\sigma_\xi^2 = 0$
$\sigma_\xi^2 = 4$
0.01
0.1
0.5
1
2
heterogeneity τ^2